

ADR-COM-16: Client Account VPC Design

Status	Approved
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Context [↗](#)

To standardize the VPC design for client accounts, we need a scalable, secure, and easily manageable network architecture. Each client account requires a dedicated VPC with isolated subnets to ensure data security and proper segmentation of resources. This design supports multiple environments, such as Dev, Test, and Prod, under predefined overarching CIDR blocks:

- Dev: 10.80.0.0/13
- Test: 10.88.0.0/13
- Prod: 10.128.0.0/9

Each VPC within these environments will use a /21 CIDR range to support scalable services like SDS migration from Oracle to AWS and future applications such as SRB.

Problem Statement [↗](#)

SDS Client accounts require unique, scalable, and non-overlapping network spaces that allow secure communication with external systems and centralized management through a Transit Gateway (TGW).

Constraints

- CIDR range: Limited to the defined overarching CIDR blocks for each environment.
- VPC must support high availability across multiple Availability Zones (AZs).
- No direct internet access from client accounts; all external connectivity must route through a dedicated network account.

Decision [↗](#)

Adopt a standardized VPC design for client accounts within the defined CIDR blocks for each environment. This design will allocate IP ranges, configure subnets, and define network connectivity according to the following specifications:

Chosen Solution:

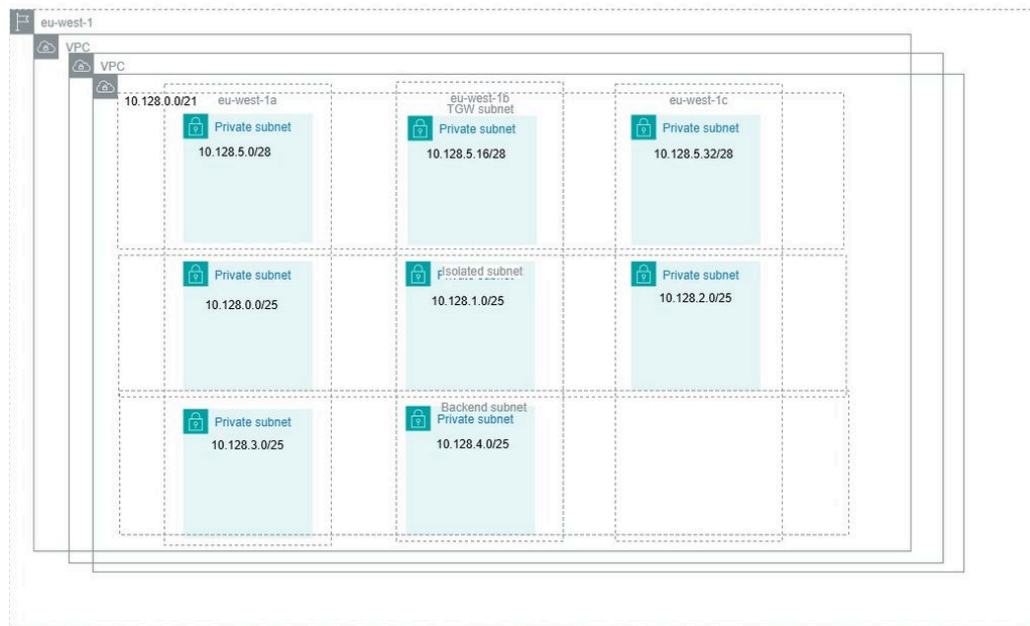
- **CIDR Allocation:**
 - Overarching CIDR for each environment:
 - Dev: 10.80.0.0/13

- Test: 10.88.0.0/13
 - Prod: 10.128.0.0/9
- Each client VPC: /21 CIDR block (2,048 IPs)
- Each subnet: /25 CIDR block (128 IPs, 126 usable)
- TGW subnets: Three /28 CIDR blocks per VPC (16 IPs each, 11 usable per subnet)
 - Selected for high availability across three AZs
 - Provides dedicated transit traffic monitoring
 - Supports compliance and audit requirements
 - Minimal impact on overall VPC address space
- **Subnet Configuration:**
 - 3 TGW subnets (one per AZ) for Transit Gateway attachments
 - 3 Isolated subnets (one per AZ) for internal resources
 - 2 Backend subnets (spread across AZs) for external-facing resources
- **Network Components:**
 - No Internet Gateway or NAT Gateway
 - Transit Gateway (TGW) attachment to enable centralized routing
 - Isolated Subnets remain isolated from TGW
- **Implementation Details:**
 - CIDR ranges tracked in DynamoDB table within `portal-shared-services`.
 - Stored as SSM Parameters in client accounts for automation.
 - Deployed using CloudFormation StackSets to standardize configuration.
 - Script provided to automate CIDR allocation for new accounts, executed by IT Support as part of their runbook.

High-Level Description

- DMZ subnets connect to TGW for communication with the network account.
- Backend subnets are isolated to protect external-facing resources.
- IPs are allocated as follows:
 - /21 VPCs provide 2,048 IPs each.
 - /25 subnets provide 128 IPs each (123 IPs usable) (5 subnets per VPC).

SDS Network Sizing



Key Configurations

- Route tables: Separate tables for Transit Gateway, Isolated and Backend subnets.
- Security Groups configured for isolation and controlled access.

Example Allocations

Examples of Allocations:

- **Dev Environment (Example for Account 1):**

- VPC CIDR: 10.80.0.0/21
 - TGW Subnets:
 - AZ1: 10.80.5.0/28
 - AZ2: 10.80.5.16/28
 - AZ3: 10.80.5.32/28
 - Isolated Subnets:
 - AZ1: 10.80.0.0/25
 - AZ2: 10.80.1.0/25
 - AZ3: 10.80.2.0/25
 - Backend Subnets:
 - AZ1: 10.80.3.0/25
 - AZ2: 10.80.4.0/25

- **Test Environment (Example for Account 1):**

- VPC CIDR: 10.88.0.0/21

- TGW Subnets:
 - AZ1: 10.88.5.0/28
 - AZ2: 10.88.5.16/28
 - AZ3: 10.88.5.32/28
- Isolated Subnets:
 - AZ1: 10.88.0.0/25
 - AZ2: 10.88.1.0/25
 - AZ3: 10.88.2.0/25
- Backend Subnets:
 - AZ1: 10.88.3.0/25
 - AZ2: 10.88.4.0/25
- **Production Environment (Example for Client Account 1):**
 - VPC CIDR: 10.128.0.0/21
 - TGW Subnets:
 - AZ1: 10.128.5.0/28
 - AZ2: 10.128.5.16/28
 - AZ3: 10.128.5.32/28
 - Isolated Subnets:
 - AZ1: 10.128.0.0/25
 - AZ2: 10.128.1.0/25
 - AZ3: 10.128.2.0/25
 - Backend Subnets:
 - AZ1: 10.128.3.0/25
 - AZ2: 10.128.4.0/25
- **Production Environment (Example for Client Account 2):**
 - VPC CIDR: 10.128.8.0/21
 - **TGW Subnets:**
 - AZ1: 10.128.13.0/28
 - AZ2: 10.128.13.16/28
 - AZ3: 10.128.13.32/28
 - Isolated Subnets:
 - AZ1: 10.128.8.0/25
 - AZ2: 10.128.9.0/25
 - AZ3: 10.128.10.0/25
 - Backend Subnets:
 - AZ1: 10.128.11.0/25

- AZ2: 10.128.12.0/25
- **Production Environment (Example for Client Account 3):**
 - VPC CIDR: 10.128.16.0/21
 - TGW Subnets:
 - AZ1: 10.128.21.0/28
 - AZ2: 10.128.21.16/28
 - AZ3: 10.128.21.32/28
 - DMZ Subnets:
 - AZ1: 10.128.16.0/25
 - AZ2: 10.128.17.0/25
 - AZ3: 10.128.18.0/25
 - Backend Subnets:
 - AZ1: 10.128.19.0/25
 - AZ2: 10.128.20.0/25

Acceptance Criteria [🔗](#)

1. A VPC is created in each client account with a /21 CIDR range within the defined overarching CIDR block for the environment.
2. CIDR ranges are stored in each account item within the DynamoDB table in portal-shared-services, passed to the CloudFormation StackSet, and saved as an SSM parameter in the client account.
3. Eight subnets are created within the VPC (3 TGW, 3 ISOLATED, 2 Backend).
4. Transit Gateway attachments are established using all three dedicated TGW subnets to connect the client VPC to the central Transit Gateway in the network account.
5. A script is provided to automate CIDR block generation for new accounts, to be run by IT Support as part of their runbook.

Consequences [🔗](#)

Positive Effects

1. **Scalability:** For future-proofing, each VPC can accommodate up to **10 additional /25 subnets**, as each VPC retains **1,360 IPs** (66.41% of the VPC space) as unallocated capacity.
2. **Security:** Segregated subnets with no direct internet access enhance security.
3. **Centralized Management:** TGW enables efficient routing and connectivity management.
4. **Automation:** Streamlines future account setup with automated CIDR allocation.
5. **Enhanced Availability:** Triple AZ TGW subnet design ensures maximum resilience.
6. **Improved Monitoring:** Dedicated TGW subnets enable precise traffic monitoring.

Negative Effects

1. **Reserved IPs:** 5 IPs reserved per subnet (for AWS use and networking) reduces usable IP space in each subnet. With 5 subnets, total of 25 IPs are reserved for AWS use.
2. **Increased Dependency:** Relying on the network account for internet access may impact client account autonomy.

Implications

- Future CIDR allocations must adhere to this structure to avoid conflicts
- TGW configurations must be regularly audited to ensure compliance and performance
- Network monitoring should include specific focus on TGW subnet utilization
- Regular testing of AZ failover scenarios is required

Related Decisions [↗](#)

None

Votes [↗](#)

Name	Vote	Comments
@Peter Drnovsek	YES	

References [↗](#)

- [AWS Transit Gateway Documentation](#)
- [AWS VPC Documentation](#)
- [CloudFormation StackSets Documentation](#)